

Karthik Narayanan

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EXECUTIVE SUMMARY

Pursuing B.S. in Data Science and Applications at IIT Madras, with strong analytical, problem-solving, and algorithm design skills built on a solid foundation in data science. GATE 2026 qualified in Data Science & AI and Computer Science & IT. Interested in pursuing M.S. and research opportunities in ML, DL, RL and AI.

EDUCATION

Indian Institute of Technology Madras (IITM)

May 2028 (Expected)

B.S. in Data Science and Applications — GPA: 7.79 / 10.0

Relevant Coursework: Deep Learning, Generative AI, Machine Learning Practice, Software Engineering, Statistics & Probability for Data Science, Linear Algebra, Data Structures & Algorithms, Database Management Systems

Sri Sairam Institute of Technology

May 2026

B.E. in Electrical and Electronics Engineering — GPA: 8.31 / 10.0

Relevant Coursework: Electric Vehicles, Renewable Energy, Smart Grid Technologies, Energy Management & Auditing, Control Systems, Power Systems, Microprocessors & Embedded Systems, Digital Electronics

PUBLICATIONS

Karthik, V.N., Rathnavel, P., Sathish Kumar, R., Jenith Joseph, A. and Palani Ganesh, H. (2025) ‘Gesture-based home automation simulation using machine learning’, in *Proceedings of the 2025 International Conference on Communication and Computational Technologies (ICCCT)*. IEEE.

<https://doi.org/10.1109/ICCCT63501.2025.11019251>

- Proposed a methodology for automating electrical appliances via hand gestures, detected and classified by a trained machine learning model, achieving 86% accuracy under controlled conditions (lighting, distance).
- Responsible for model programming and training.

RESEARCH PROJECTS

Robust Music Genre Classification under Noisy Mashup Conditions

- Built for the *Messy Mashup* competition, evaluating genre-classification robustness under realistic, noisy mixing conditions (tempo-synced, same-genre stem mashups with randomized noise) rather than clean training tracks, across 10 genres and 4 instrument stems (drums, vocals, bass, other).
- Converted stems to spectrograms and fine-tuned AST and HuBERT models alongside a custom CNN (`torch.nn`) with a Squeeze-and-Excitation attention mechanism, achieving 92% accuracy on the noisy mashup test set.
- Tools:* Python, PyTorch, scikit-learn, matplotlib, seaborn

Comment Category Prediction

- Built for the *Comment Category Prediction Challenge*: predicted the final category assigned to user comments from interaction feedback, symbolic expressions, topic-reference indicators, and system signals (textual, numerical, categorical features; anonymized integer labels).
- Engineered features across all three modalities; trained and compared Logistic Regression, LightGBM, and XGBoost with hyperparameter tuning, reaching 82% accuracy with the best model.
- Tools:* Python, scikit-learn, LightGBM, XGBoost, matplotlib, seaborn

Household Services Application

- Designed and built a multi-user full-stack platform (admin, service professionals, customers) for booking and managing home-servicing solutions, connecting residents with vetted local providers; used Redis and Celery for asynchronous background task processing (automated daily reminders, monthly activity reports) without blocking the main request cycle.
- Tools:* Python, Flask, Flask-SQLAlchemy (SQLite), Vue.js, Redis, Celery, HTML & CSS, Chart.js

TECHNICAL SKILLS

Programming Languages Python, C/C++, Java, JavaScript, SQL, Dart

ML & Deep Learning PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn, XGBoost, LightGBM, CNN, Transfer Learning, LSTM, RNN, GAN

Data & Backend Tools Pandas, NumPy, matplotlib, Flask, Redis, Celery, Vue.js

Research Interests Distributed & Scalable ML Systems, Autonomous Systems, Computer Vision, Algorithm Optimization, MLOps, Audio & Multimodal Deep Learning, Full-Stack Data Applications

HONORS & ACHIEVEMENTS

- Qualified the Graduate Aptitude Test in Engineering (GATE) 2026 in **Data Science & Artificial Intelligence** and **Computer Science & Information Technology**.